

Exercise 6.10

An electron in a hydrogen atom is in the energy eigenstate

$$\psi_{2,1,-1}(r, \theta, \varphi) = N r e^{-r/2a_0} Y_{1,-1}(\theta, \varphi).$$

- (a) Find the normalization constant, N .
- (b) What is the probability per unit volume of finding the electron at $r = a_0$, $\theta = 45^\circ$, $\varphi = 60^\circ$?
- (c) What is the probability per unit radial interval (dr) of finding the electron at $r = 2a_0$? (One must take an integral over θ and φ at $r = 2a_0$.)
- (d) If the measurements of $\hat{L}^2$ and $\hat{L}_z$ were carried out, what will be the results?

* Here $\psi_{2,1,-1} = u_{2,1,-1}$
This is just a notational difference.

Recall: $u_{n\ell m} = R_{n\ell}(r) Y_{\ell m}(\theta, \varphi)$
So

$$u_{2,1,-1} = R_{2,1}(r) Y_{1,-1}(\theta, \varphi)$$

Ⓞ There are two ways to solve this.
* The first, we simply read N off of the table for $R_{2,1}(r)$

$$R_{2,1}(r) = 3^{-1/2} (2a_0)^{-3/2} \frac{r}{a_0} e^{-r/2a_0}$$

Plugging this into $u_{2,1,-1}$:

$$u_{2,1,-1} = \left(3^{-1/2} (2a_0)^{-3/2} a_0^{-1} \right) r e^{-r/2a_0} Y_{1,-1}(\theta, \varphi)$$

so

$$N = 3^{-1/2} (2a_0)^{-3/2} a_0^{-1} = \frac{1}{2a_0^2 \sqrt{6a_0}}$$

* The Second way is by brute force:
we know that

$$\langle n, \ell, m | n, \ell, m \rangle = 1$$

$$\Rightarrow \int \int \int |u_{n\ell m}|^2 r^2 \sin\theta dr d\theta d\varphi = 1$$

$$= \iiint |N r e^{-r/2a_0}|^2 |Y_{1,-1}(\theta, \phi)|^2 r^2 \sin(\theta) dr d\theta d\phi$$

$$= N^2 \left(\int_0^\infty r^4 e^{-r/a_0} dr \right) \underbrace{\left(\int_0^{2\pi} \int_0^\pi |Y_{1,-1}(\theta, \phi)|^2 \sin(\theta) d\theta d\phi \right)}_{=1}$$

$$= N^2 \int_0^\infty r^4 e^{-r/a_0} dr$$

Recall from problem 2:

$$u = r/a_0 \Rightarrow a_0 du = dr$$

$$\Rightarrow = N^2 \int_0^\infty (a_0 u)^4 e^{-u} (a_0 du)$$

$$= N^2 a_0^5 \int_0^\infty u^4 e^{-u} du$$

$$= N^2 a_0^5 4! = N^2 a_0^5 4 \cdot 3 \cdot 2 \cdot 1 = N^2 a_0^5 24$$

Therefore

$$N = \frac{1}{\sqrt{a_0^5 24}} = \frac{1}{2a_0^2 \sqrt{6a_0}}$$

just as above.

Gamma function: $\Gamma(n+1) = n! = \int_0^\infty u^n e^{-u} du$

(You could also do integration by parts... but that's gross...)

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Recall that the probability of finding the state in a small volume dV where $dV = r^2 \sin(\theta) dr d\theta d\phi$

is given by

$$|\psi_{n\ell m}|^2 dV$$

therefore,

$|\psi_{n\ell m}|^2$: Probability per unit volume

So we simply plug in the above values

$$\psi_{21,-1}(a_0, 45^\circ, 60^\circ) = \left(\frac{1}{2a_0^3 \sqrt{6a_0}} \right) a_0 e^{-r/2a_0} Y_{1,-1}(45, 60)$$

→ look up in our table for $Y_{1,-1}$

$$Y_{1,-1}(\theta, \phi) = \sqrt{\frac{3}{8\pi}} \sin(\theta) e^{-i\phi}$$

$$45^\circ = \pi/4$$

$$60^\circ = \pi/3$$

$$\begin{aligned} Y_{1,-1}(45, 60) &= \sqrt{\frac{3}{8\pi}} \sin\left(\frac{\pi}{4}\right) e^{-i\pi/3} \\ &= \sqrt{\frac{3}{8\pi}} \left(\frac{1}{\sqrt{2}}\right) e^{-i\pi/3} \end{aligned}$$

$$|\psi_{21,-1}|^2 = \left(\frac{1}{4a_0^4 (6a_0)} \right) (a_0^2 e^{-1}) \left(\left(\frac{3}{8\pi}\right) \left(\frac{1}{2}\right) \right)$$

$$|\psi_{21,-1}|^2 = \frac{1}{128\pi e a_0^3}$$

Since a_0 is the Bohr radius, this has units of $\frac{1}{\text{Volume}} \sim \frac{1}{r^3}$ as desired.

Ⓒ Recall from problem 1 that the radial prob. dist. was given by

$$\begin{aligned} P_{21}(r) &= r^2 |R_{21}(r)|^2 \\ &= r^2 \left(\frac{1}{3 \cdot 8 \cdot a_0^3} \cdot \frac{r^2}{a_0^2} \cdot e^{-r/a_0} \right) \\ &= \frac{1}{24 a_0^5} r^4 e^{-r/a_0} \end{aligned}$$

• We want to know this value at $r = 2a_0$

$$P_{21}(2a_0) = \frac{1}{24 a_0 e^2}$$

Ⓓ Straight forward.

